

Technical Requirements Document (TRD) — EcoStep

Product: EcoStep

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Audience: Engineering, QA, DevOps, Security, Data, Product Design

1. Purpose

This Technical Requirements Document translates the EcoStep PRD into an implementable engineering specification. It defines the target architecture, system components, data model, APIs, security controls, performance requirements, testing scope, release gates, and operational requirements needed to deliver the MVP / Soft Launch and the Full V1.0 launch.

2. Scope

In Scope

- Social authentication onboarding
- Lifestyle archetype-based footprint baseline generation
- Region-based emissions factor lookup
- Dual-Value Dashboard
- Manual eco-action logging
- Behavioral AI Micro-Habits engine
- Local community challenges and league enrollment
- Rewards points ledger and Eco-Perks readiness
- Analytics, observability, security, and privacy controls
- Admin and configuration tooling for emissions and equivalency datasets

Out of Scope for MVP

- Utility account integrations
- Precise geolocation history storage
- Full smart-home device integrations (Nest, Ecobee)
- Real-money payments system for rewards settlement
- International support beyond initial configured markets unless explicitly enabled

3. Product-to-Technical Mapping

PRD Requirement	Technical Translation
Baseline profile in under 60 seconds	Lightweight onboarding flow, precomputed emissions factors, minimal data capture, async profile enrichment
Carbon + cash dashboard	Aggregation service combining emissions, savings, habits, and equivalency conversions
3 personalized daily habits	Rules + ranking engine with capped recommendation count and completion feedback loop
Community challenges	Geo-bucketed league service, leaderboard aggregation, anti-fraud scoring
Privacy-first design	Data minimization, consent store, pseudonymized analytics, revocable permissions
50,000 concurrent users	Horizontally scalable stateless APIs, caching, queue-based async workloads

4. System Overview

EcoStep will use a service-oriented cloud architecture optimized for mobile clients and web clients. The system should separate latency-sensitive user interactions from asynchronous computation such as habit generation, leaderboard refreshes, cohort analytics, and localized equivalency updates.

Core Services

1. Client Apps

- iOS / Android mobile app
- Responsive web application

2. API Gateway

- Authentication, request routing, throttling, versioning

3. Identity Service

- Apple / Google OAuth
- User session and token lifecycle

4. User Profile Service

- Archetype, region, preferences, consent state

5. Carbon Calculation Service

- Baseline estimation
- Action-to-impact conversion
- Regional emissions factor management

6. Savings & Equivalency Service

- Cash savings estimates
- Relatable equivalency conversions

7. Habits Personalization Engine

- Daily generation of exactly 3 recommendations

- Completion feedback processing

8. Challenges & Rewards Service

- Geo-leagues, points, rewards redemption hooks

9. Analytics & Experimentation Service

- Funnel tracking, retention, feature flags, A/B tests

10. Admin Console

- Dataset management, challenge setup, moderation, audit review

5. Recommended Architecture

5.1 High-Level Stack

- **Frontend:** React Native for mobile, React / Next.js or equivalent for web admin and web client
- **Backend:** Node.js / TypeScript or Java / Kotlin microservices
- **Database:** PostgreSQL for transactional data
- **Cache:** Redis for sessions, rate limiting, leaderboard caching, hot dashboard reads
- **Search / Analytics Store:** BigQuery / ClickHouse / Snowflake equivalent for event analytics
- **Object Storage:** S3-compatible storage for static assets, challenge images, exported reports
- **Message Queue:** Kafka / PubSub / SQS + workers for asynchronous jobs
- **Observability:** OpenTelemetry, centralized logging, metrics, tracing, alerting
- **Infrastructure:** Kubernetes or managed container platform with autoscaling

5.2 Architectural Principles

- Stateless APIs for horizontal scaling
- Event-driven processing for expensive or bursty workloads
- Dataset versioning for emissions and equivalency registries
- Region-aware configuration to support localized calculations
- Strict separation of PII from behavioral analytics events
- Graceful degradation when recommendation or challenge services are delayed

6. Functional Technical Requirements

6.1 Onboarding Service

Input

- Social auth identity token
- Selected lifestyle archetype
- Zip / postal code
- Optional notification consent

Processing

- Validate social token
- Create user account and profile shell
- Resolve region to city / metro / emissions profile
- Assign default league bucket
- Generate baseline monthly footprint estimate
- Initialize streaks, points ledger, and dashboard cache

Output

- Authenticated session
- Baseline carbon score
- Estimated monthly savings opportunity band
- Initial 3 micro-habits

Technical Acceptance Criteria

- End-to-end onboarding API latency p95 under 800 ms excluding auth provider callback
- No more than 4 required user interactions after sign-in
- Failed zip/postal resolution must fall back to city/state selection without blocking account creation

6.2 Dual-Value Dashboard

Required Data Blocks

- Current monthly carbon score
- Comparison with local average
- Dollar savings to date (month, quarter, lifetime)
- Daily streak / habits completion
- Equivalency translation cards
- Challenge standing and points balance

Technical Requirements

- Dashboard aggregation endpoint should assemble data from cache-first read models
- Dashboard first meaningful response p95 under 1.5 seconds on standard 4G-equivalent network conditions
- Cached dashboard snapshot refresh interval no greater than 5 minutes for non-critical widgets
- Manual refresh support for habits, points, and challenge status

6.3 Carbon Calculation Engine

Responsibilities

- Generate baseline estimate from archetype + region + optional household modifiers
- Convert logged actions into CO2e avoided / reduced values
- Support versioned regional factors (EPA, EEA, future registries)
- Store provenance metadata for each calculation version

Rules

- Every calculation must reference a `factor_version_id`
- Historical user metrics must remain reproducible even after factor updates
- Recalculation jobs must be idempotent and auditable

6.4 Savings Estimation Engine

Responsibilities

- Estimate financial savings attached to eco-actions
- Support user-visible confidence bands where precision is low
- Separate estimated savings from verified savings in the schema and UI

Example Categories

- Electricity reduction
- Reduced food waste
- Lower transport spend
- Appliance usage optimization

6.5 Behavioral AI Action Engine

Core Requirement

The system must deliver exactly 3 daily micro-habits per user.

Inputs

- User archetype
- Region
- Habit completion history
- Time of day / local time zone
- Active challenge context
- Optional user preference exclusions

Output Constraints

- Exactly 3 habits
- No duplicate habit IDs in a day
- At least 1 habit must be completable in under 30 seconds
- At least 2 habits must align with the user archetype category score

Suggested Technical Approach

- Phase 1: rules engine + ranking model
- Phase 2: contextual bandit / ML ranking with safety constraints
- Fallback path: curated archetype habit packs if personalization service is unavailable

6.6 Community Challenges & Geo-Leagues

Capabilities

- Auto-enroll users into city/neighborhood league
- Support campaign-based challenges by geography and category
- Generate leaderboards by individuals and teams
- Detect suspicious logging patterns

Requirements

- League assignment based on coarse region only, not precise GPS history by default
- Leaderboard refresh cadence: every 15 minutes for MVP, near real time for V1.0 if cost permits

- Challenge scoring rules must be versioned and configurable by admin

6.7 Rewards & Eco-Perks

MVP

- Points accrual ledger
- Reward catalog placeholder / mock inventory support
- Redemption reservation workflow for test market partners

V1.0

- Marketplace integration APIs
- Coupon / voucher generation
- Inventory and expiration handling
- Fraud and abuse checks

7. Data Model

7.1 Core Entities

- users
- user_consent
- user_profiles
- regions
- emission_factor_versions
- baseline_profiles
- eco_actions
- user_action_logs
- habit_definitions
- daily_habit_recommendations
- habit_completions
- leagues
- challenge_definitions
- challenge_participation
- leaderboard_snapshots
- points_ledger
- reward_catalog
- reward_redemptions

- `equivalency_definitions`
- `audit_logs`

7.2 Sample Schema Notes

users

- `user_id` UUID PK
- `auth_provider`
- `provider_subject_id`
- `created_at`
- `status`

user_profiles

- `user_id` FK
- `display_name`
- `archetype`
- `region_id`
- `household_size_optional`
- `timezone`
- `privacy_mode`

baseline_profiles

- `baseline_id`
- `user_id`
- `factor_version_id`
- `estimated_monthly_co2e`
- `estimated_monthly_savings_low`
- `estimated_monthly_savings_high`
- `calculated_at`

daily_habit_recommendations

- `recommendation_id`
- `user_id`
- `date_local`
- `habit_1_id`
- `habit_2_id`

- habit_3_id
- generation_strategy
- generated_at

points_ledger

- entry_id
- user_id
- source_type
- source_ref_id
- points_delta
- balance_after
- created_at

8. API Requirements

8.1 External / Client APIs

- POST /v1/auth/social/login
- POST /v1/onboarding/complete
- GET /v1/dashboard
- POST /v1/actions/log
- GET /v1/habits/today
- POST /v1/habits/{id}/complete
- GET /v1/challenges
- GET /v1/leaderboards/{leagueId}
- GET /v1/rewards/catalog
- POST /v1/rewards/redeem
- GET /v1/profile
- PATCH /v1/profile
- POST /v1/consents
- DELETE /v1/consents/{type}

8.2 API Standards

- REST/JSON for client-facing APIs
- Idempotency keys required for write endpoints with retry risk
- OAuth2 / OIDC tokens for user auth
- Pagination for leaderboard, rewards catalog, and history feeds

- Structured error model with user-safe messages and internal correlation IDs

9. Security & Privacy Requirements

9.1 Data Minimization

- No storage of third-party utility credentials
- No precise geolocation history stored unless explicit consent is granted
- Postal code should be normalized into coarse region IDs for ongoing analytics where possible
- PII and behavioral event data stored in separate logical domains

9.2 Compliance Controls

- GDPR and CCPA rights workflows: export, deletion, consent revocation, purpose limitation
- Data retention policies by data class
- Audit logging for admin actions and sensitive data access
- Vendor and subprocessors inventory maintenance

9.3 Security Controls

- Encryption in transit via TLS 1.2+
- Encryption at rest for databases, backups, object storage
- Secrets in managed secret store
- RBAC for internal admin tooling
- Periodic dependency and container vulnerability scans
- WAF, bot mitigation, API rate limiting
- Mandatory third-party penetration and privacy audits before V1.0

9.4 PII Leak Prevention

- DLP scanning in logs and analytics pipelines
- Redaction middleware for sensitive fields
- Prohibit raw provider identity tokens in logs
- Security incident alerting with sev classification and on-call workflow

10. Non-Functional Requirements

10.1 Performance

- Support 50,000 concurrent users during peak notification windows
- Dashboard load p95 under 1.5 seconds under standard 4G conditions
- API availability target: 99.9% monthly for MVP, 99.95% for V1.0
- Push-triggered habit fetch p95 under 700 ms from edge/API gateway to response start

10.2 Reliability

- Graceful degradation for non-core features such as marketplace and challenge detail graphics
- Retry-safe job processing with dead-letter queues
- RPO under 15 minutes, RTO under 1 hour for critical services

10.3 Scalability

- Horizontal pod autoscaling on CPU, memory, and queue depth
- Read replicas and cache offload for leaderboard and dashboard traffic
- Batch pre-generation of daily habit recommendations before local morning peak

10.4 Localization

- Support region-specific emissions registries
- Unit conversions for metric / imperial display
- Currency formatting by locale
- Time zone-aware daily habit and streak calculations

11. Analytics & Experimentation

Core Events

- Sign-up started / completed
- Onboarding step completion
- Dashboard viewed
- Habit shown / completed / dismissed
- Action logged
- Challenge joined / progressed
- Reward viewed / redeemed
- Consent granted / revoked

Required Metrics

- Onboarding drop-off rate
- Day-1, Day-7, Day-30 retention
- Habit completion rate
- Average estimated CO2e reduction per active user
- Average estimated savings per active user
- Challenge participation rate
- Crash-free sessions

Experimentation Support

- Feature flags by geography and cohort
- A/B tests for dashboard wording, equivalency types, and habit framing
- Event schema governance with versioning

12. QA & Test Strategy

12.1 Test Layers

- Unit tests for calculation services and recommendation rules
- Contract tests for APIs
- Integration tests for auth, onboarding, dashboard aggregation, ledger updates
- E2E tests for key user journeys on Android, iOS, and web
- Performance tests for 50,000 concurrent user target
- Security tests including SAST, DAST, dependency scans, pen test

12.2 Critical Test Scenarios

1. User completes onboarding in under 60 seconds flow path
2. Invalid or unknown postal code fallback behavior
3. Daily habit generation always returns exactly 3 items
4. Dashboard values remain consistent after factor version updates
5. Duplicate habit completion taps do not double-award points
6. League assignment works without precise geolocation storage
7. Consent revocation removes non-essential data processing
8. Rewards redemption prevents double spending

12.3 Release Gate Validation

- Crash rate under 0.1%
- Main dashboard load under 1.5 seconds on target network profile
- Zero confirmed PII leaks in third-party audit findings

13. DevOps & Operations

CI/CD

- Trunk-based development with protected main branch
- Automated unit, integration, security, and lint checks
- Blue/green or canary deploys for backend services
- Mobile feature flagging to decouple deployment from release

Observability

- SLIs/SLOs for auth, dashboard, habits, challenge services
- Distributed tracing across gateway, services, queue workers, and DB
- Centralized dashboards for latency, error rates, queue lag, and notification send success

Incident Management

- Sev1 to Sev4 classification
- On-call rotation
- Runbooks for auth outage, cache degradation, queue backlog, and data sync issues

14. Release Plan

Phase 1: MVP / Soft Launch (Q3)

Engineering Scope

- Social auth
- Archetype + region onboarding
- Baseline calculator v1
- Manual action logging
- Dashboard v1
- Rules-based 3-habit engine
- Single-market geo-leagues
- Points ledger
- Core analytics and observability

Market Scope

- One metro area, such as Austin, TX
- Curated challenge set and limited reward partners

Phase 2: Full V1.0 Global Launch (Q4)

Additional Scope

- Smart-home integrations
- Rewards marketplace APIs
- Regional/city leaderboards
- Expanded localization coverage
- Enhanced fraud controls
- ML-based habit ranking improvements

15. Risks & Mitigations

Risk	Impact	Mitigation
Inaccurate savings estimates reduce trust	High	Show ranges/confidence bands, disclose estimated values, validate with pilot data
Regional emissions data quality varies	High	Version datasets, document source provenance, regional fallback hierarchy
Manual logging fraud inflates challenges	Medium	Rate limits, anomaly scoring, caps, audit review, delayed reward release
Over-personalization increases privacy risk	High	Minimize inputs, separate PII, explicit consent gates, coarse region defaults
Peak notification traffic causes dashboard latency	High	Precompute read models, autoscaling, cache hot paths, queue smoothing

16. Open Questions

- What level of household customization is acceptable in MVP without increasing onboarding drop-off?
- Should equivalency messaging be globally standardized or regionally localized by culture?
- What partner integration standard will be used for Eco-Perks in non-US markets?
- Will challenge anti-fraud actions be fully automated or require manual moderation thresholds?
- Which emissions datasets are the official source of truth for launch geographies?

17. Definition of Done

A release is considered technically complete only when:

- All scoped features meet functional acceptance criteria
- SLO and load-test thresholds are met
- Security, privacy, and compliance checks pass
- Observability and alerting are active in production
- Support and incident runbooks are approved
- Analytics instrumentation is validated end to end
- Product, engineering, QA, and security sign-off is documented